

Question	Answer	Marks
4(a)	<u>sum / total</u> momentum (of a system of bodies) is constant or <u>sum / total</u> momentum before = <u>sum / total</u> momentum after	M1
	for an isolated system / no (resultant) <u>external</u> force	A1
4(b)(i)	$p = mv$ or 4.0×6.0 or 2.0×3.0 $(4.0 \times 6.0) + (2.0 \times 3.0) = 6.0 \times v$	C1
	so $v = 5.0 \text{ m s}^{-1}$	A1

Question	Answer	Marks
4(b)(ii)	KE before $= (0.5 \times 4.0 \times 6.0^2) + (0.5 \times 2.0 \times 3.0^2)$ $= 81 \text{ J}$	C1
	KE after $= (0.5 \times 6.0 \times 5.0^2)$ $= 75 \text{ J}$	(C1)
	percentage transferred $= [(81 - 75) / 81] \times 100$ $= 7\%$	A1

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